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<https://swayam.gov.in>https://swayam.gov.in/nc_details/NPTEL

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NPTEL (<https://swayam.gov.in/explorer?ncCode=NPTEL>) » Introduction To Machine Learning - IITKGP
(course)



Course outline

About
NPTEL ()

How does an
NPTEL
online
course
work? ()

Week 0 ()

Week 1 ()

Week 2 ()

Week 3 ()

Week 4 ()

Week 5 ()

- ☐ Lecture 23 :
Logistic
Regression
(unit?
unit=40&lesso
n=41)

Week 5 : Assignment 5

The due date for submitting this assignment has passed.

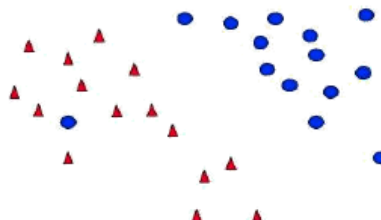
Due on 2024-08-28, 23:59 IST.

Assignment submitted on 2024-08-28, 22:17 IST

1)

2 points

What would be the ideal complexity of the curve which can be used for separating the two classes shown in the image below?



- A) Linear
B) Quadratic
C) Cubic
D) insufficient data to draw conclusion

- ☒ A)
☐ B)
☐ C)
☐ D)

Yes, the answer is correct.
Score: 2

Accepted Answers:
A)

2)

2 points

☐ Lecture 24:
Introduction
Support Vector
Machine (unit?
unit=40&lesso
n=42)

☐ Lecture 25:
SVM : The
Dual
Formulation
(unit?
unit=40&lesso
n=43)

☐ Lecture 26:
SVM :
Maximum
Margin with
Noise (unit?
unit=40&lesso
n=44)

☐ Lecture 27:
Nonlinear
SVM and
Kernel
Function (unit?
unit=40&lesso
n=45)

☐ Lecture 28:
SVM : Solution
to the Dual
Problem (unit?
unit=40&lesso
n=46)

☐ Lecture 29:
Python
Exercise on
SVM (unit?
unit=40&lesso
n=47)

☐ Week 5 :
Lecture
Material (unit?
unit=40&lesso
n=48)

☒ Quiz: Week 5
: Assignment
5
(assessment?
name=146)

☐ Feedback
Form for Week

Suppose you have a dataset with $n=10$ features and $m=1000$ examples. After training a logistic regression classifier with gradient descent, you find that it has high training error and does not achieve the desired performance on training and validation sets. Which of the following might be promising steps to take?

1. Use SVM with a non-linear kernel function
2. Reduce the number of training examples
3. Create or add new polynomial features

- A) 1, 2
B) 1, 3
C) 1, 2, 3
D) None

☐ A)

☒ B)

☐ C)

☐ D)

Yes, the answer is correct.

Score: 2

Accepted Answers:

B)

3)

2 points

In logistic regression, we learn the conditional distribution $p(y|x)$, where y is the class label and x is a data point. If $h(x)$ is the output of the logistic regression classifier for an input x , then $p(y|x)$ equals:

- A. $h(x)^y(1 - h(x))^{(1-y)}$
B. $h(x)^y(1 + h(x))^{(1-y)}$
C. $h(x)^{1-y}(1 - h(x))^y$
D. $h(x)^y(1 + h(x))^{(1+y)}$

☒ A.

☐ B.

☐ C.

☐ D.

Yes, the answer is correct.

Score: 2

Accepted Answers:

A.

4)

2 points

The output of binary class logistic regression lies in the range:

- A. $[-1, 0]$
B. $[0, 1]$
C. $[-1, -2]$
D. $[1, 10]$

☐ A.

☒ B.

5 (unit?
unit=40&lesso
n=127)

☐ Assignment 5
Solution (unit?
unit=40&lesso
n=156)

Week 6 ()

Week 7 ()

Week 8 ()

Assignment
Solution ()

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Videos ()

Problem
Solving
Session -
July 2024 ()

☐ C.

☐ D.

Yes, the answer is correct.

Score: 2

Accepted Answers:

B.

5)

2 points

State whether True or False.

"After training an SVM, we can discard all examples which are not support vectors and can still classify new examples."

A) TRUE

B) FALSE

☒ A)

☐ B)

Yes, the answer is correct.

Score: 2

Accepted Answers:

A)

6)

2 points

Suppose you are dealing with a 3-class classification problem and you want to train a SVM model on the data. For that you are using the One-vs-all method. How many times do we need to train our SVM model in such a case?

A) 1

B) 2

C) 3

D) 4

☐ A)

☐ B)

☒ C)

☐ D)

Yes, the answer is correct.

Score: 2

Accepted Answers:

C)

7)

2 points

What is/are true about kernels in SVM?

1. Kernel function can map low dimensional data to high dimensional space
2. It's a similarity function

A) 1

B) 2

C) 1 and 2

D) None of these.

☐ A)



- ☐ B)
- ☒ C)
- ☐ D)

Yes, the answer is correct.

Score: 2

Accepted Answers:

C)

8)

2 points

If $g(z)$ is the sigmoid function, then its derivative with respect to z may be written in term of $g(z)$ as

- A) $g(z)(g(z)-1)$
- B) $g(z)(1+g(z))$
- C) $-g(z)(1+g(z))$
- D) $g(z)(1-g(z))$

- ☐ A)
- ☐ B)
- ☐ C)
- ☒ D)

Yes, the answer is correct.

Score: 2

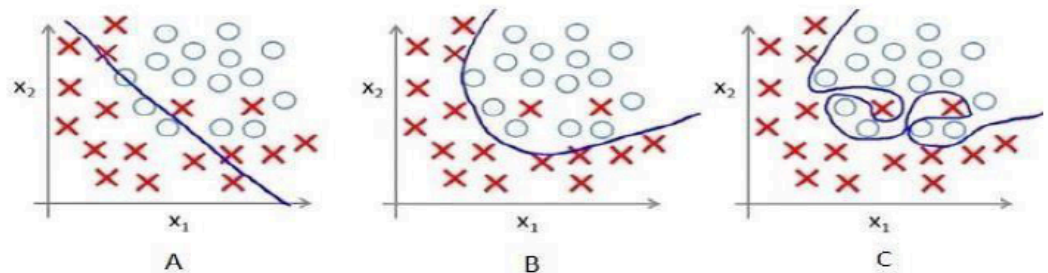
Accepted Answers:

D)

9)

2 points

Below are the labelled instances of 2 classes and hand drawn decision boundaries for logistic regression. Which of the following figures demonstrates overfitting of the training data?



- A) A
- B) B
- C) C
- D) None of these

- ☐ A)
- ☐ B)
- ☒ C)
- ☐ D)

Yes, the answer is correct.

Score: 2

Accepted Answers:

C)

10)

2 points

What do you conclude after seeing the visualization in the previous question (Question 9)?

- C1. The training error in the first plot is higher as compared to the second and third plot.
- C2. The best model for this regression problem is the last (third) plot because it has minimum training error (zero).
- C3. Out of the 3 models, the second model is expected to perform best on unseen data.
- C4. All will perform similarly because we have not seen the test data.

- A) C1 and C2
- B) C1 and C3
- C) C2 and C3
- D) C4

- ☐ A)
- ☒ B)
- ☐ C)
- ☐ D)

Yes, the answer is correct.

Score: 2

Accepted Answers:

B)

